

AUSELIA

Analog Front-End (AFE) v1: bring-up guide

From the first look at the board to the sensor on the plant.

Each step has a goal, a list of actions, a table of expected values, and a check list for a wrong result. The simulation of the as-built board gives each expected value.

Version 0.1 / November 4, 2026 / Nicolás Villegas

1 About this guide

This guide takes the first AFE boards from the box to a measured result. The steps follow the signal. First, apply power. Then test the chain, the listener, and the SPI (serial peripheral interface) link. Last, test the sensor head and put it on the plant.

Each step has the same parts: a goal, a list of actions, a table of expected values, a green Pass box, and an amber If-not box. The simulation of the as-built netlist gives the expected values. No real board has given a measurement yet. Your readings are the first.

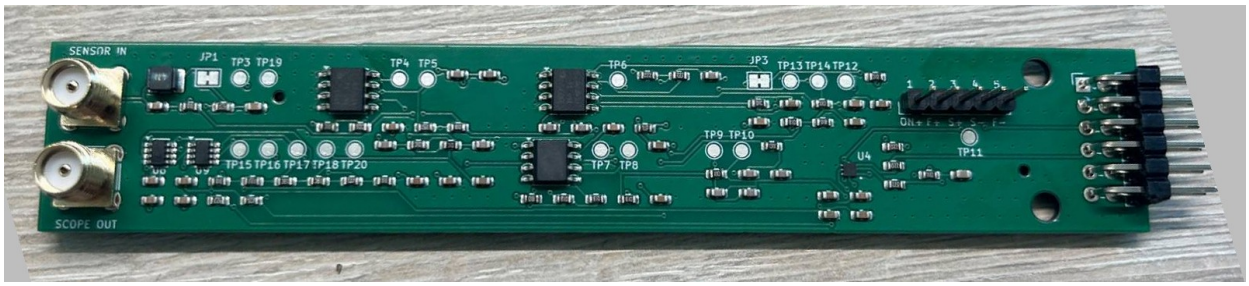


Figure 1. The AFE v1 test board, top side. SENSOR IN (J1) and SCOPE OUT (J4) are on the left. The power header J7 and the PMOD (peripheral module connector) J3 are on the right.

1.1 What you need

Item	Use
Keysight B2901BL source-measure unit (SMU)	Supplies power through J7 and measures the supply current with four wires
Function generator or Analog Discovery 2	Makes bursts of 10 cycles at 250 kHz from a 50 Ω output
Oscilloscope, 250 MS/s or faster	Measures the test points with a 1 MΩ input
Multimeter	Measures resistance and DC voltage
ZedBoard with a program that reads the ADC	Runs the SPI link and reads the listener lines (Steps 6 and 7)
10 pF and 1600 pF C0G capacitors	Inject charge and replace the sensor
SMA cables, a jumper for J7, a knife, a soldering iron	Cut and close JP1 and JP3
Magnifier or microscope	Examines U4, U6, and U9

1.2 Test setup

Figure 2 shows the setup for the signal tests in Steps 3 to 5.

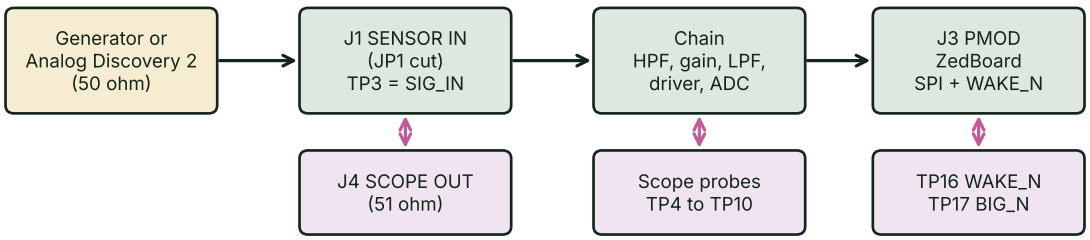


Figure 2. Test setup for Steps 3 to 5.

Cut JP1 before you connect a generator to J1. The center pin of J1 carries 3.3 V DC when JP1 is closed. The generator would receive this voltage. Cut JP1 with a knife for Steps 3 to 5. Close JP1 again with solder for Steps 8 to 10.

Touch a grounded surface before you touch the board. The ADC U4 and the comparators U6 and U9 are small parts with a fine pitch.

2 Know the board

Figure 3 shows each part and each test point (TP). The red dot on an IC marks pin 1. Table 1 lists each test point and the stage it measures.

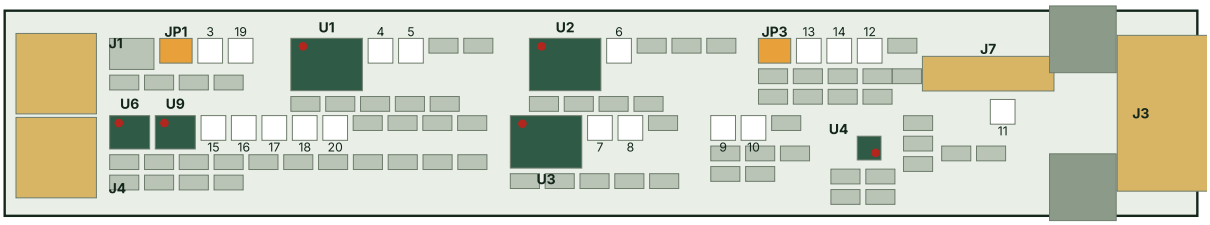


Figure 3. Top view with each part. Green: ICs with a red pin 1 dot. Gold: connectors. Orange: solder jumpers JP1 and JP3. White: test points, shown by number.

Table 1. Test points of the main board. SIG_IN is the input after the bias-T.

TP	Net	Shows	TP	Net	Shows
TP3	SIG_IN	input	TP12	AVDD	ADC reference
TP4	HPF_OUT	high-pass filter	TP13	VMID5	1.65 V mid-rail
TP5	GAIN_OUT	gain stage	TP14	VOCM	driver center
TP6	LPF_OUT	low-pass filter	TP15	LST	listener signal
TP7	DRV_OUTP	driver +	TP16	WAKE_N	small window
TP8	DRV_OUTN	driver −	TP17	BIG_N	large window
TP9	AINP	ADC input +	TP18	REFL	listener center
TP10	AINM	ADC input −	TP19, TP20	GND	scope ground
TP11	+3V3	board rail			

2.1 The signal path

Figure 4 shows the signal size at each test point for a sine at 250 kHz. The gain of the chain to the ADC is about 4.1. The driver divides the signal into two halves of opposite sign around 1.65 V.

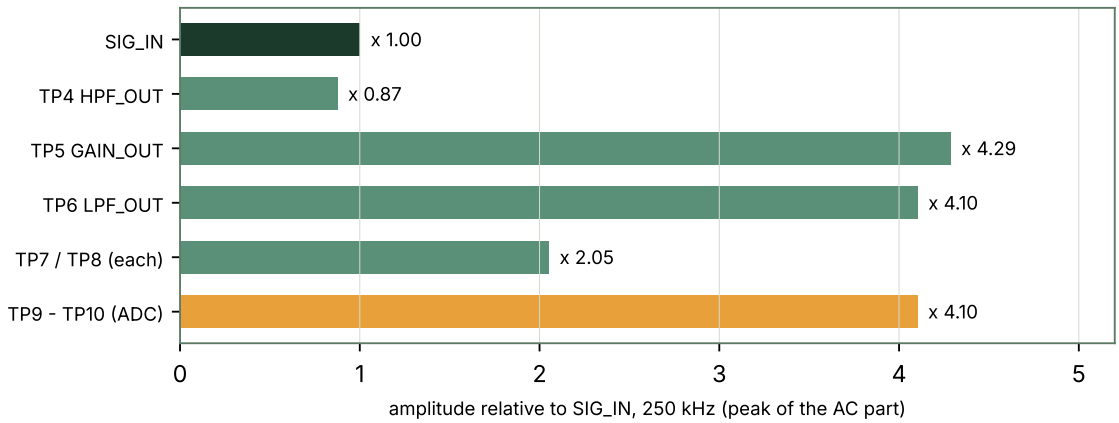


Figure 4. Expected amplitude at each test point at 250 kHz, relative to SIG_IN (ngspice, as-built netlist).

STEP 0 Look first, with the power off

Goal: find a wrong part, a short, or a solder bridge before current flows.

1. Examine the top side with the magnifier.
2. Compare the board with Figure 3 and with the photo on page 2.
3. Make sure pin 1 of each IC is at the correct end. Use Table 2.
4. Examine U4, U6, and U9 for solder bridges between pads.
5. Make sure the pins of J3 and J7 are straight.
6. Make sure J3 is flush with the board edge.
7. Measure the resistance from `TP11` (+3V3) to `TP19` (GND). Wait until the value is stable.

Table 2. Pin 1 positions. View from the top, with SENSOR IN on the left.

Ref	Part	Package	Pin 1
U1, U2, U3	OPA2365, OPA2365, THS4531	SOIC-8	top left, next to the silkscreen triangle
U4	ADS7057	X2QFN-8	lower right of the part
U6, U9	TLV7022	SOT-23-8	top left
U8 (head)	OPA365	SOT-23-5	lower pin on the right side, next to the silkscreen triangle

The resistance is about 2 kΩ. This is an estimate: the VOCM divider (2 kΩ), the VMID5 divider (20 kΩ), and the ladder (2.1 MΩ) are in parallel. Any value above 1 kΩ is correct. No bridges are visible. Each pin 1 agrees with Table 2.

If the resistance is below 500 Ω, there is a short. Do not apply power. Look for a bridge at U4, U6, or U9. Look for a turned IC or a bridged C0G capacitor. A value that rises slowly is only the capacitors that charge.

STEP 1 Power up with the SMU

Goal: make sure the board uses the correct current and each rail has its value.

J7 is the power entry. Do not connect the ZedBoard in this step.

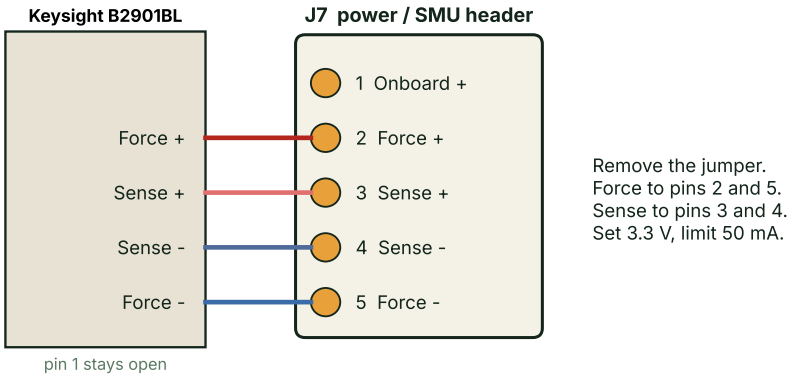


Figure 5. Four-wire connection of the B2901BL to J7. Pins 2 and 3 meet at the board rail. Pins 4 and 5 are ground.

1. Remove the jumper from J7. Do not connect the PMOD. Leave JP1 closed.
2. Set the SMU to voltage source mode. Set the voltage to 3.3 V. Set the current limit to 50 mA. Turn remote sense on. Keep the output off.
3. Connect Force to pins 2 and 5 of J7. Connect Sense to pins 3 and 4.

Turn the output off if the current is more than 60 mA. This prevents damage to the board.

4. Turn the output on. Read the current after one second.
5. Measure each node in Table 3 against GND (TP19).

Table 3. Expected DC values (ngspice, all links closed, head not connected). The tolerance is ±30 mV for 3.3 V nodes and ±50 mV for other nodes.

Node	Test point	Expected	Note
+3V3	TP11	3.30 V	the SMU value
AVDD	TP12	3.27 V	after the 10 Ω and 10 µF filter
VMID5	TP13	1.65 V	needs 25 ms after power-up
VOCM	TP14	1.64 V	half of AVDD
LST	TP15	1.64 V	rests at REFL
WAKE_N, BIG_N	TP16, TP17	3.30 V	open drain with pull-up
REFL	TP18	1.64 V	middle of the ladder
Supply current	SMU	20 to 22 mA	about 25 mA with the head

The current is 18 to 30 mA. Each node agrees with Table 3.

If the current is too high, turn the power off and do Step 2. If TP11 is low, check the Sense wires. If a mid-rail is wrong, wait 25 ms and measure again. Then check R7, R8, R14, and R15.

STEP 2 Split the current with JP3

Goal: find which part of the board uses the current.

JP3 connects the capture rail to the 3.3 V rail. The capture rail supplies U1, U2, U3, and AVDD. The listener and DVDD are on the 3.3 V rail.

1. Turn the power off.
2. Cut JP3 with the knife.
3. Connect the SMU to the capture side of the gap, or put a second ammeter across the gap.
4. Measure the current of each side.
5. Close JP3 with a drop of solder.

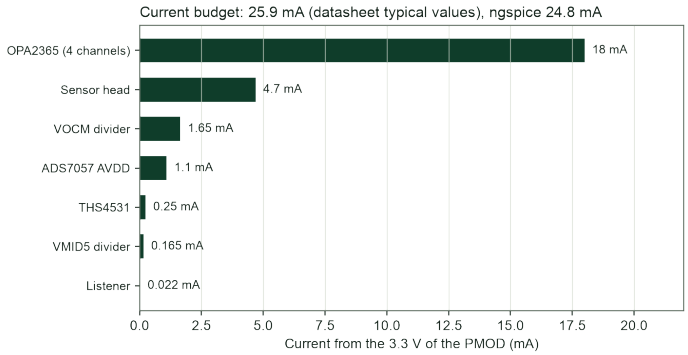


Figure 6. Current budget (datasheet typical values, estimate). The four op-amp channels use the most current.

The capture rail uses 20 to 22 mA. The other side uses about 0.3 mA.

If the capture side is high, check U1, U2, U3, and the capacitors near them. If the other side is high, check U4, U6, and U9.

STEP 3 The signal chain at 250 kHz

Goal: make sure each stage has the gain that the simulation gives.

Cut JP1 before you connect the generator. Keep the amplitude at SIG_IN below 300 mV peak. A larger signal clips the gain stage.

1. Apply power as in Step 1.
2. Set the generator to a burst of 10 cycles at 250 kHz. Set the repeat time to 1 ms. Set the amplitude to 100 mV peak.
3. Connect the generator to J1.
4. Trigger the oscilloscope on TP3 (SIG_IN).
5. Measure the peak of the AC part at each test point in Table 4.
6. Look at TP7 and TP8. Make sure they have opposite phase around 1.65 V (Figure 7).
7. Measure SCOPE OUT (J4) with a 1 MΩ input. Make sure it agrees with TP6.

Table 4. Expected peak at each test point for 100 mV peak at SIG_IN and 250 kHz. The limit is ±10 %.

Test point	Net	Peak (mV)	Gain
TP3	SIG_IN	100	1.00
TP4	HPF_OUT	87	0.87
TP5	GAIN_OUT	429	4.29
TP6	LPF_OUT	410	4.10
TP7, TP8	DRV_OUTP, DRV_OUTN	±205 each	opposite signs around 1.65 V
TP9, TP10	AINP, AINM	±205 each	same as the driver
TP15	LST	68	0.68 (listener filter)

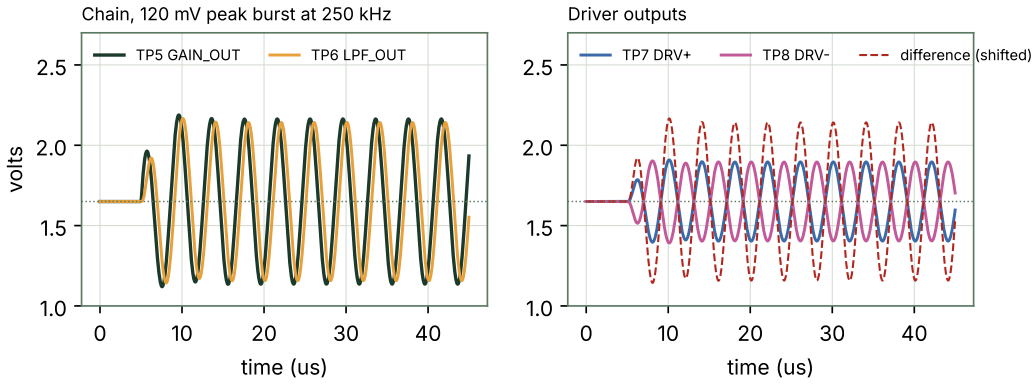


Figure 7. Expected waveforms for a burst of 120 mV peak (ngspice). The ADC measures the difference of TP7 and TP8.

Each peak is within 10 % of Table 4.

Find the first stage that is wrong. If TP4 is wrong, check C1, C2, R1, R2, and pin 1 of U1. If TP5 is wrong, check R3 (3.9 kΩ) and R4 (1 kΩ). If TP6 is wrong, check R5, R6, and C4 to C6. If TP7 and TP8 are not opposite, check U3, R10 to R13, and VOCM at TP14.

STEP 4 The frequency response

Goal: make sure the band agrees with the simulation.

1. Set the generator to a continuous sine of 100 mV peak.
2. Set the frequency to 10 kHz. Read the peak at TP6.
3. Repeat for 30 kHz, 100 kHz, 250 kHz, 500 kHz, 1 MHz, and 2 MHz.
4. Calculate the gain of each frequency relative to 250 kHz, in dB.
5. Compare the result with Table 5.

Table 5. Expected response relative to 250 kHz (ngspice, whole chain to the ADC). The limit is ±1.5 dB from 100 kHz to 500 kHz and ±3 dB at other frequencies.

Frequency	10 kHz	30 kHz	100 kHz	250 kHz	500 kHz	1 MHz	2 MHz
Gain (dB)	−40.5	−21.5	−3.1	0	−2.8	−12.1	−24.0

The result is within the limit of Table 5 at each frequency.

An error at the low end points to C1, C2, R1, or R2. An error at the high end points to C4, C5, C6, R5, or R6.

STEP 5 The listener: WAKE_N and BIG_N

Goal: make sure the two window comparators fire near 35 mV and 111 mV at SIG_IN.

Keep the setup of Step 3, with JP1 cut. Probe TP15 (LST), TP16 (WAKE_N), TP17 (BIG_N), and TP18 (REFL).

1. Make sure TP18 reads 1.65 V.
2. Make sure TP16 and TP17 read 3.3 V with no signal.
3. Send the burst at 20 mV peak at SIG_IN. Note the state of TP16 and TP17.
4. Send the burst at 60 mV peak. Note the state of TP16 and TP17.
5. Send the burst at 150 mV peak. Note the state of TP16 and TP17.
6. To find the thresholds, increase the amplitude in steps of 5 mV. Note the amplitude when WAKE_N falls and when BIG_N falls.

A line stays low only for the length of the burst.

Table 6. Expected lines for a burst of 10 cycles at 250 kHz, with JP1 cut. The amplitude is the peak at SIG_IN. Each threshold has a tolerance of ±12 mV.

Amplitude	WAKE_N (TP16)	BIG_N (TP17)
20 mV	stays high	stays high
60 mV	falls	stays high
150 mV	falls	falls
Threshold	about 35 mV	about 111 mV

Each line agrees with the table. Each threshold is within ±12 mV. The response time is about 260 ns.

If WAKE_N is always low, make sure TP18 is 1.65 V. Then check R39 to R41, R21, and R23. If a threshold is far from the table, check R22, R41 (33 kΩ), R39, and R40 (15 kΩ). If no line falls, check U6, U9, R24, and R42.

STEP 6 The SPI link

Goal: make sure the SPI frames are correct.

The ZedBoard must run a program that drives CS and SCLK and reads SDO. Table 7 gives the pins. The ADC only sends data, so the link has no MOSI line.

Power from the ZedBoard

Do this part first. The SPI link needs the ZedBoard.

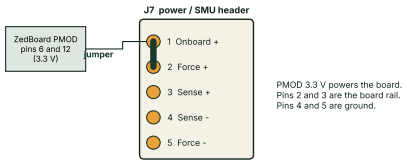


Figure 8. Power from the ZedBoard PMOD with a jumper on J7 pins 1 and 2.

1. Turn the power off.
2. Put the jumper on J7 pins 1 and 2.

> Do not use the jumper and the SMU at the same time. The two sources would connect together.

3. Connect the board to a PMOD port of the ZedBoard.

Read the frames

Step 6, continued. Use the program on the ZedBoard.

Table 7. Pins of the PMOD J3. The board uses nine of the twelve pins.

Pin	Signal	Pin	Signal
1	CS	7	BIG_N
2	not used	8	not used
3	SDO	9	not used
4	SCLK	10	WAKE_N
5	GND	11	GND
6	3V3	12	3V3

1. Apply no signal to J1. Set SCLK to 10 MHz.
2. Send one frame of 24 clocks. This frame calibrates the offset.
3. Send frames of 18 clocks.
4. Look at CS, SCLK, and SDO at the J3 pins. Compare them with Figure 9.
5. Make sure the first frame returns zeros. Make sure each next frame returns the sample of the frame before.

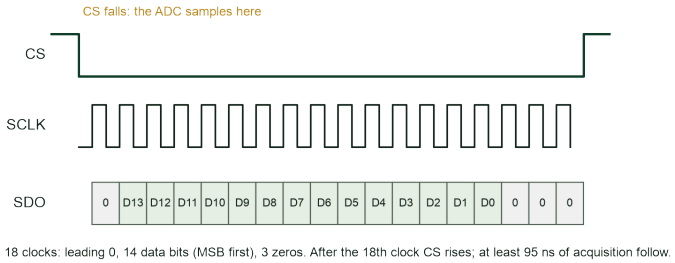


Figure 9. One frame. CS falls and the ADC samples. After 18 clocks, CS rises and the ADC acquires for at least 95 ns.

CS, SCLK, and SDO agree with Figure 9. The first frame returns zeros. Each next frame returns the sample of the frame before.

If all data is zero, read the data twice, because the data comes one frame late. If SDO is stuck, check +3V3 at pin 4 of U4 and R18 to R20 (22 Ω).

STEP 7 The ADC codes

Goal: make sure the codes follow the input.

1. Apply no signal to J1. Read 4096 samples. Calculate the mean (offset) and the rms noise in codes. One code is 0.403 mV.
2. Apply a sine of 100 mV peak at 250 kHz at SIG_IN. Read 4096 samples. Find the peak code.
3. Increase SCLK to 20 MHz, then to 25 MHz. Look for errors in the samples.
4. If the samples have errors, move the sample edge of the host later.

Table 8. Expected codes.

Test	Expected	Limit
First frame (24 clocks)	all zeros	-
Offset with the input idle	about 0 codes	±10 codes
Noise with the input idle (no sensor)	about 1 code rms (estimate)	below 3 codes
Peak for 100 mV at SIG_IN and 250 kHz	±1018 codes	±10 %
Sample rate at 10, 20, 25, and 60 MHz	0.53, 1.01, 1.23, and 2.50 MSPS	-

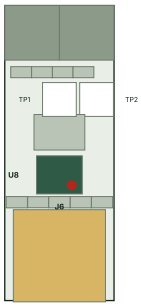
The codes follow the input in size and sign. SCLK at 25 MHz gives no errors. The idle noise is below 3 codes.

If the word is wrong, remember the format: one leading zero, 14 data bits (D13 first, two's complement), and three zeros. If the offset is large, check VOCM at TP14 and VMID5 at TP13.

STEP 8 The sensor head alone

Goal: make sure the charge amplifier converts charge to voltage with the gain Q/Cf.

The head (10 × 27 mm) gets power through the coax from SENSOR IN. Close JP1 on the main board. Connect an SMA cable from J6 of the head to J1.



Head connections

TP1 (+)	piezo + pad (summing node)
TP2 (−)	piezo − pad (head ground)
J6	SMA to J1 of the main board
U8	OPA365, pin 1 as in Step 0

Figure 10. The sensor head. The two large holes at the top are the M2 mounting holes.

1. Wait 25 s. The reference divider (two 1 MΩ resistors and 10 µF) needs this time to settle.
2. Measure the DC voltage at the head. The pad of C37 must be about 2.9 V. The pad of C35 must be about 1.45 V.
3. Make sure the supply current is about 4.3 mA higher than in Step 1.
4. Connect the generator through a 10 pF C0G capacitor to TP1. Connect the generator ground to TP2.
5. Set the generator to a burst of 10 cycles at 250 kHz.
6. Read the peak at TP3 on the main board for each row of Table 9.
7. Stop the generator. Connect the 1600 pF C0G capacitor between TP1 and TP2 as a dummy sensor.
8. Record TP6 with the input quiet. The noise gain is 35, as with the real sensor.

Table 9. Expected head output for Cf = 47 pF and Q = 10 pF × V. The tolerance is ±10 %.

Generator (peak)	Charge	Head output at TP3	TP6 (chain)	Listener
100 mV	1 pC	21 mV	87 mV	no change
300 mV	3 pC	64 mV	262 mV	WAKE_N falls
1000 mV	10 pC	213 mV	874 mV	both fall

Each output is within 10 % of Table 9. A real event of 4 to 7 pC gives 85 to 149 mV. WAKE_N falls for each event, and BIG_N falls above 5.2 pC. The noise at the ADC with the dummy sensor is below 2 codes. The simulation gives 0.2 codes.

If there is no output, make sure JP1 is closed and the head has DC power. If the DC voltage is wrong, check pin 1 of U8, R29, and R30. If the gain is wrong, check C34 (47 pF C0G) and R28 (1 MΩ). If the noise is very high with the dummy sensor, wait 25 s. Then check the coax and the ground.

STEP 9 The head and the board together

Goal: make sure one full chain works from the head connector to the ZedBoard.

1. Connect the head to J1 with the coax. Leave JP1 closed.
2. Power the board from the PMOD.
3. Inject 3 pC as in Step 8 while the ZedBoard reads the ADC.
4. Find the peak code.
5. Make sure WAKE_N falls for the length of the burst.

The peak code is about 650. This value is 262 mV divided by 0.403 mV. WAKE_N falls for the length of the burst.

STEP 10 The sensor on the plant

Goal: make the first real recording.

1. Solder the piezo sensor to the head.
2. Mount the sensor on the plant.
3. Record a window that is long enough for several events.
4. Compare the peak frequency and the amplitude with the perfboard results. The old data has peaks at 233 to 275 kHz and charges of 4 to 7 pC.
5. Note the rms noise in codes.
6. Note the highest frequency in an event.

The two last numbers help the next design.

3 Troubleshooting

Start at the top of Figure 11. Find the first question with the answer no. Look only at that stage.

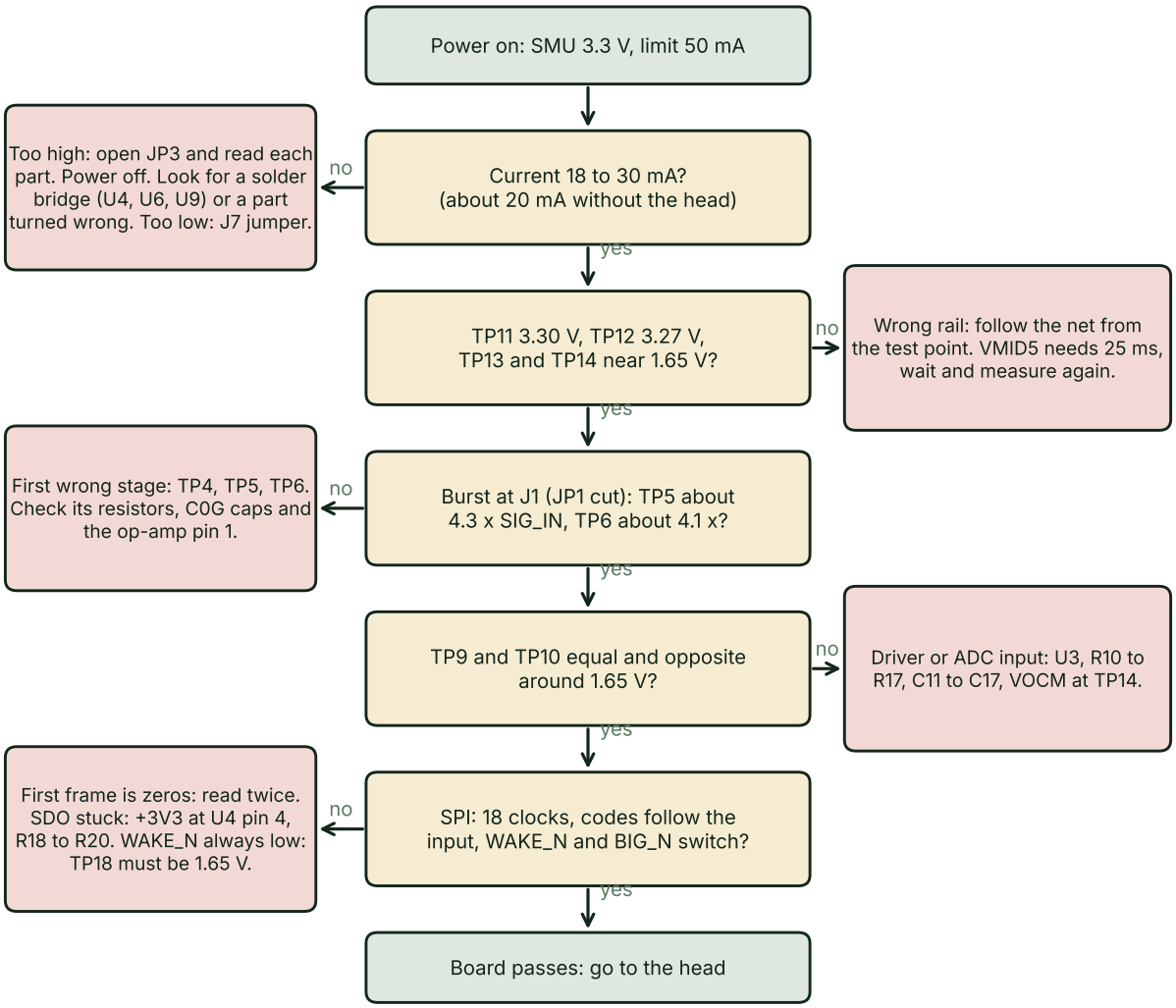


Figure 11. Decision chart for the main board.

4 Record sheet

Write the readings in the table. Board serial: _____ Date: _____ Power: ☐ SMU ☐ PMOD

Check	Expected	Measured	Limit	OK
TP11 to TP19 resistance	about 2 kΩ		$>$ 1 kΩ	
Supply current (no head)	20 to 22 mA		18 to 30 mA	
+3V3 (TP11)	3.30 V		±30 mV	
AVDD (TP12)	3.27 V		±50 mV	
VMID5 (TP13)	1.65 V		±50 mV	
VOCM (TP14)	1.64 V		±50 mV	
REFL (TP18)	1.64 V		±50 mV	
TP5 / TP3 at 250 kHz	4.29		±10 %	
TP6 / TP3 at 250 kHz	4.10		±10 %	
TP7, TP8 (each, 100 mV in)	±205 mV		±10 %	
Gain at 100 / 500 kHz (dB)	-3.1 / -2.8		±1.5 dB	
WAKE_N threshold	35 mV		±12 mV	
BIG_N threshold	111 mV		±12 mV	
ADC offset (codes)	0		±10	
ADC noise, input idle (codes rms)	about 1		$<$ 3	
Peak code for 100 mV at 250 kHz	1018		±10 %	
Head: DC at C37 / C35	2.9 V / 1.45 V		±0.1 V	
Head: 3 pC at TP3	64 mV		±10 %	

Notes